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SVInsight: A Python Package for Calculating Social Vulnerability Indices

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Summary

The exposure of communities to environmental hazards, their sensitivity to such events, and their ability to respond (adaptive capacity) are influenced by their social, political, cultural, economic, and demographic information (Susan L. Cutter et al., 2010; Fatemi et al., 2017; Smit & Wandel, 2006). Understanding the interconnected relationships among exposure, sensitivity, and adaptive capacity is important to estimate the degree to which stakeholders can mitigate environmental hazards (Smit & Wandel, 2006). Social Vulnerability Indices, or SVIs, are built on social and demographic data to serve as proxies for these interconnected variables. Numerous national and international SVIs exist including, SoVI ® from The University of South Carolina's Hazards Vulnerability & Resilience Institute (Susan L. Cutter et al., 2003), the Center for Disease Control (Flanagan et al., 2011), and the United Nations Development Program (UNDP, 2010), as well as more region-specific indicators, such as Texas A&M's Hazard Reduction and Recovery Center's Coastal Indicator (Peacock et al., 2010) and the Austin Area Sustainability Indicator (Bixler et al., 2021). In this paper, we present an open-source Python package, *SVInsight*, which provides an accessible workflow to calculate social vulnerability indices that are specific to user-defined study areas while using the two leading calculation methodologies.

Statement of need

Generic social vulnerability indices for large regions can be lacking in their ability to identify at risk populations (Nelson et al., 2015; Tate, 2012; Tellman et al., 2020). Furthermore, not all vulnerability indices are created equal, and users, researchers, and developers should clearly state the objectives and structure of their index in order to accurately present their findings (Bakkensen et al., 2016). Research utilizing SVIs has historically been limited to either existing national or regional databases (Bixler et al., 2021; Susan L. Cutter et al., 2003; Flanagan et al., 2011; Van Zandt et al., 2012). Researchers that want to incorporate social vulnerability information into their research are therefore limited to these existing databases and their pre-determined variable choices, or developing their own index from scratch. Furthermore, researchers' understanding of what contributes to vulnerability is variable in both time and space (Susan L. Cutter & Finch, 2008). Therefore, it is necessary to tailor SVIs to the context in which they are being applied.

SVInsight is an accessible and open-source tool to quickly calculate SVIs for a user-defined region using either custom or a research supported pre-set list of social and demographic variables. This package creates a pipeline between extensively large datasets and easily customizable SVIs, allowing researchers to experiment and manipulate various indices (e.g., general vulnerability, economic, race/ethnicity, etc.) built on a variety of variables more

efficiently and effectively.*SVInsight* calculates SVIs built on data from the Census Bureau's 5-Year American Community Survey using the two leading methods for calculating such metrics: a composite score using a data reduction methodology (Susan L. Cutter et al., 2003) and a ranking method (Flanagan et al., 2011). There is a recently published package in R that is capable of calculating the Rank Method SVI (Xu et al., 2024), but to the best of the author's knowledge there is no open source factor analysis based SVI package.

Background

SVInsight was developed to provide an open-source tool for computing social vulnerability indices (SVIs) based on user-defined study areas and input variables. SVInsight uses two methodologies to create indices based on existing scholarly research: (1) a factor analysis method and (2) a rank method. This tool's purpose is to provide researchers with the ability to quickly create exploratory estimates of vulnerability based on existing and known relationships between socio-demographic variables and a community's sensitivity (a proxy for vulnerability). Indices created with SVInsight can be used as a starting point for understanding the complex relationship between socio-demographic. Furthermore, it is important to note that vulnerability estimates are relative, and calculated scores should be interpreted in the context of the study area being examined.

The *Factor Analysis* method is influenced by the original SoVI®, which utilizes a principal component analysis (Susan L. Cutter et al., 2003). By utilizing a data reduction methodology, a large subset of American Community Survey 5-Year Estimate variables can be dimensionally reduced and combined into a single index. A principal component analysis (PCA) reduces data by creating one or more index variables, or 'components,' using a linear combination (i.e., a weighted average) from a larger set of measured variables. The purpose of a PCA is to determine the optimal number of components, the optimal choice of measured variables for each component, and the optimal weights. A factor analysis is a model of latent variables, which are underlying or unobservable constructs that are believed to be responsible for the patterns of correlations among observed variables. We leverage a factor analysis approach because it is the data reduction methodology most often associated with creating composite indices. To summarize the factor analysis methodology: (1) standardize input variables from 0 to 1; (2) conduct initial factor analysis with maximum number of possible factors; (3) calculate initial eigenvalues to determine significant and appropriate number of factors; (4) recompute factor analysis with new number of factors; (5) calculate the loading factors to determine significant variables; (6) calculate variance statistics to determine cumulative sum of variance captured in the composite score; and (7) begin iterative factor analysis until optimal number of factors with only significant variables is determined and return the final SVI values.

The *Rank Method* was popularized by the Center for Disease Control's SVI, and is built on the rank or percentile of each input variable (Flanagan et al., 2011). To summarize the rank method: (1) sort each variable from high to low and give a rank value; (2) sum the ranks for each variable in each location; and (3) locations with higher overall rank have higher vulnerability score. In most cases, the factor analysis and rank methods will look similar and have the same general geographic patterns of vulnerability, such as the case for Travis County, Texas, where the clear distinction between the western and eastern halves of the county is visible (Figure 1). Both methods are relevant and used in various research settings as work continues to be done to test the validity and applicability of SVIs across fields of research (Rufat et al., 2019; Spielman et al., 2020; Tate et al., 2021).

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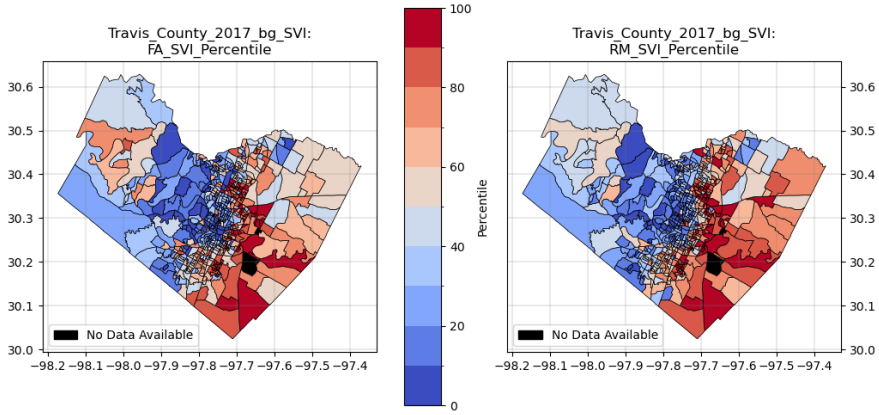

Figure 1: Example output from SVInsight showing the 2017 factor analysis (left) and rank method (right) social vulnerability index estimates for Travis County, Texas.

Functionality and Ease of Use

SVInsight automates the SVI creation process. The entire workflow can be completed in only 5 lines of code:

- `project = svi(project_name, file_path, api_key, geoids)`: Initializes an SVI project and creates the necessary file structure for all of the intermediary files that might be of interest to users
- `project.boundaries_data(boundary, year)`: Extracts the appropriate Census geographic boundaries for the project area
- `project.census_data(boundary, year)`: Extracts the raw Census data and fills NaN values (e.g., no Census reported information for a specific location and year) as needed
- `project.configure_variables(config_file)`: Configures the specific SVI run
- `project.calculate_svi(config_file, boundary, year)`: Calculates both SVI estimates (Factor Analysis and Rank Methods)

In addition, there are methods to include/exclude additional variables to test different sets of inputs into the SVI calculation. An example jupyter notebook as well as sample python scripts are provided to showcase how *SVInsight* functions at different scales under different scenarios.

Projects Using *SVInsight*

The work done leading up to the creation of the *SVInsight* Python package was and continues to be a part of numerous vulnerability and hazard related research efforts. The origination of this project started with the creation of a methodology to support urban community resilience to hazards in Austin, Texas (Bixler et al., 2021). Since then, it became a tool for incorporating social vulnerability information in a household flood impact model (Preisser et al., 2022) and a near-real-time critical resource accessibility model (Preisser et al., 2023). *SVInsight* continues to be utilized in various hazard impact studies across Texas.

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Author Contributions

Matthew Preisser: Conceptulization (equal), Methodology (equal), Software (lead), Visualization (lead), Writing - original draft (lead), Writing - review & editing (lead); Paola Passalacqua: Conceptulization (equal), Methodology (equal), Software (supporting) Writing - review & editing (supporting), Supervision (lead); R. Patrick Bixler: Conceptulization (equal), Methodology (equal), Writing – review and editing (supporting), Supervision (supporting).

References

Bakkensen, L. A., Fox-Lent, C., Read, L. K., & Linkov, I. (2016). Validating resilience and vulnerability indices in the context of natural disasters. *Risk Analysis*, *37*(5), 982–1004. https://doi.org/10.1111/risa.12677

Bixler, R. P., Yang, E., Richter, S. M., & Coudert, M. (2021). Boundary crossing for urban community resilience: A social vulnerability and multi-hazard approach in austin, texas, USA. *International Journal of Disaster Risk Reduction*, *66*, 102613. https://doi.org/10.1016/j.ijdrr.2021.102613

Cutter, Susan L., Boruff, B. J., & Shirley, W. L. (2003). Social vulnerability to environmental hazards. *Social Science Quarterly*, *84*(2), 242–261. https://doi.org/10.1111/1540-6237.8402002

Cutter, Susan L., Burton, C. G., & Emrich, C. T. (2010). Disaster resilience indicators for benchmarking baseline conditions. *Journal of Homeland Security and Emergency Management*, *7*(1). https://doi.org/10.2202/1547-7355.1732

Cutter, Susan L., & Finch, C. (2008). Temporal and spatial changes in social vulnerability to natural hazards. *Proceedings of the National Academy of Sciences*, *105*(7), 2301–2306.

Fatemi, F., Ardalan, A., Aguirre, B., Mansouri, N., & Mohammadfam, I. (2017). Social vulnerability indicators in disasters: Findings from a systematic review. *International Journal of Disaster Risk Reduction*, *22*, 219–227. https://doi.org/10.1016/j.ijdrr.2016.09.006

Flanagan, B. E., Gregory, E. W., Hallisey, E. J., Heitgerd, J. L., & Lewis, B. (2011). A social vulnerability index for disaster management. *Journal of Homeland Security and Emergency Management*, *8*(1). https://doi.org/10.2202/1547-7355.1792

Nelson, K. S., Abkowitz, M. D., & Camp, J. V. (2015). A method for creating high resolution maps of social vulnerability in the context of environmental hazards. *Applied Geography*, *63*, 89–100. https://doi.org/10.1016/j.apgeog.2015.06.011

Peacock, W. G., Brody, S., Seitz, W. A., Merrell, W. J., Vedlitz, A., Zahran, S., Harriss, R. C., & Stickney, R. R. (2010). Advancing the resilience of coastal localities: Developing, implementing and sustaining the use of coastal resilience indicators. *Hazard Reduction and Recovery Center*. https://doi.org/10.13140/RG.2.2.35146.80324

Preisser, M., Passalacqua, P., Bixler, R. P., & Boyles, S. (2023). A network-based analysis of critical resource accessibility during floods. *Frontiers in Water*, *5*. https://doi.org/10.3389/frwa.2023.1278205

Preisser, M., Passalacqua, P., Bixler, R. P., & Hofmann, J. (2022). Intersecting near-real time fluvial and pluvial inundation estimates with sociodemographic vulnerability to quantify a

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household flood impact index. *Hydrology and Earth System Sciences*, *26*(15), 3941–3964. https://doi.org/10.5194/hess-26-3941-2022

Rufat, S., Tate, E., Emrich, C. T., & Antolini, F. (2019). How valid are social vulnerability models? *Annals of the American Association of Geographers*, *109*(4), 1131–1153. https://doi.org/10.1080/24694452.2018.1535887

Smit, B., & Wandel, J. (2006). Adaptation, adaptive capacity and vulnerability. *Global Environmental Change*, *16*(3), 282–292. https://doi.org/10.1016/j.gloenvcha.2006.03.008

Spielman, S. E., Tuccillo, J., Folch, D. C., Schweikert, A., Davies, R., Wood, N., & Tate, E. (2020). Evaluating social vulnerability indicators: Criteria and their application to the social vulnerability index. *Natural Hazards*, *100*(1), 417–436. https://doi.org/10.1007/s11069-019-03820-z

Tate, E. (2012). Uncertainty analysis for a social vulnerability index. *Annals of the Association of American Geographers*, *103*(3), 526–543. https://doi.org/10.1080/00045608.2012.700616

Tate, E., Rahman, M. A., Emrich, C. T., & Sampson, C. C. (2021). Flood exposure and social vulnerability in the United States. *Natural Hazards*, *106*(1), 435–457. https://doi.org/10.1007/s11069-020-04470-2

Tellman, B., Schank, C., Schwarz, B., Howe, P. D., & Sherbinin, A. de. (2020). Using disaster outcomes to validate components of social vulnerability to floods: Flood deaths and property damage across the USA. *Sustainability*, *12*(15), 6006. https://doi.org/10.3390/su12156006

UNDP. (2010). *Human Development Report 2010: The Real Wealth of Nations - Pathways to Human Development*. (p. 905). https://doi.org/10.2307/2137795

Van Zandt, S., Peacock, W. G., Henry, D. W., Grover, H., Highfield, W. E., & Brody, S. D. (2012). Mapping social vulnerability to enhance housing and neighborhood resilience. *Housing Policy Debate*, *22*(1), 29–55.

Xu, H., Li, R., & Bilal, U. (2024). findSVI: An r package to calculate the social vulnerability index at multiple geographical levels. *Journal of Open Source Software*, *9*(99), 6525.

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